

NÉRON MODELS ON SMOOTH FORMAL SCHEMES

ABSTRACT. We study Néron models over a complete discrete valuation ring with algebraically closed residue field and fraction field of characteristic zero. We give an existence criterion whose hypothesis concerns pointed analytic maps from open polydiscs and whose conclusion is a mapping property on all smooth formal schemes locally formally of finite type. The representing group is π -adic and locally topologically of finite type. We also prove a criterion for lifting analytic maps from their restrictions to completed local neighbourhoods. Canonical group smoothening preserves maps from this entire category of tests. Under explicit uniformization hypotheses, these results give formal Néron models for commutative analytic groups, identify their identity completions and integral Lie algebras, and yield an alternative construction of algebraic Néron models of abelian varieties.

1. INTRODUCTION

Let R be a complete discrete valuation ring with uniformizer π , fraction field K of characteristic zero, and algebraically closed residue field k . An algebraic Néron model of a smooth separated algebraic K -group A is a smooth separated R -scheme $\mathcal{A}$, locally of finite type, with generic fibre A and

$$\mathrm{Map}_R(X, \mathcal{A}) \simeq \mathrm{Map}_K(X_K, A)$$

for every smooth R -scheme X locally of finite type. We construct it through an extension problem on smooth formal schemes.

Let SmForm_R be the category of smooth formal R -schemes locally formally of finite type. Smoothness means formal smoothness, and the finite-type condition means that the reduced scheme is locally of finite type over k . Morphisms are formal R -morphisms, and covers are adic étale covers. Write $\mathrm{SmForm}_R^{\mathrm{ad}}$ for the full subcategory of π -adic objects. It consists of the smooth π -adic formal schemes locally topologically of finite type.

These categories are related by the local structure theorem for smooth formal morphisms: every object of SmForm_R is locally the completion of an object of $\mathrm{SmForm}_R^{\mathrm{ad}}$ along a closed subscheme of its special fibre. The centre of completion need not be smooth. Write

$$\mathbb{D}_R^m = \mathrm{Spf}_{(\pi, T)} R[[T_1, \dots, T_m]], \quad D_K^m = (\mathbb{D}_R^m)_\eta$$

for the formal and analytic open polydiscs, pointed by $T_1 = \dots = T_m = 0$.

Let An_K be the category of Huber adic spaces locally of finite type over $\mathrm{Spa}(K, R)$. The subscript η denotes the adic generic fibre over R . For a smooth separated analytic K -group G , define

$$\mathcal{F}_G : \mathrm{SmForm}_R^{\mathrm{op}} \longrightarrow \mathrm{Spaces}, \quad \mathfrak{X} \longmapsto \mathrm{Map}_K(\mathfrak{X}_\eta, G). \quad (1)$$

Analytic étale descent makes this a sheaf. We seek a separated representative that belongs to the smaller category $\mathrm{SmForm}_R^{\mathrm{ad}}$. Its restriction to $\mathrm{SmForm}_R^{\mathrm{ad}}$ is the usual formal Néron model.

Theorem A (Polydisc criterion). *Let G be a smooth separated analytic K -group. Suppose $\mathfrak{P} \in \text{SmForm}_R^{\text{ad}}$ is a separated quasi-compact formal group equipped with an open immersion of groups $i: \mathfrak{P}_\eta \hookrightarrow G$. Assume that for every $m \geq 0$, generic fibre induces an equivalence*

$$\text{Map}_{R,*}(\mathbb{D}_R^m, \mathfrak{P}) \xrightarrow{\sim} \text{Map}_{K,*}(D_K^m, G). \quad (2)$$

Then there is a separated formal group $\mathfrak{N} \in \text{SmForm}_R^{\text{ad}}$ and an open immersion $j: \mathfrak{N}_\eta \hookrightarrow G$ inducing

$$\text{Map}_R(\mathfrak{X}, \mathfrak{N}) \xrightarrow{\sim} \text{Map}_K(\mathfrak{X}_\eta, G) \quad (\mathfrak{X} \in \text{SmForm}_R). \quad (3)$$

On this entire site,

$$h_{\mathfrak{N}} \simeq (\underline{G(K)} \times h_{\mathfrak{P}}) // \underline{\mathfrak{P}(R)}. \quad (4)$$

The model is quasi-compact if and only if $G(K)/\mathfrak{P}(R)$ is finite. Neither commutativity nor quasi-compactness of G is required.

Here $//$ denotes the colimit of the indicated action, and underlining denotes the constant sheaf. For a subgroup $H \subset J$ of discrete groups, $J/H := J//H$ uses right translation. Pointed maps in (2) send the origin section to the identity; the maps in (3) are arbitrary.

The additional extension step is the following result, which applies without a group structure on the target.

Theorem B (Extension from completed neighbourhoods). *Let $\mathfrak{X} \in \text{SmForm}_R$, let $\mathfrak{N}$ be an R -flat separated π -adic formal R -scheme locally topologically of finite type, and let $j: \mathfrak{N}_\eta \hookrightarrow V$ be an open immersion into a separated analytic K -space. For an analytic map $f: \mathfrak{X}_\eta \rightarrow V$, there exists a morphism $g: \mathfrak{X} \rightarrow \mathfrak{N}$ satisfying $j \circ g_\eta = f$ if and only if, for every closed $x \in \mathfrak{X}_{\text{red}}$, there exists $\phi_x: \widehat{\mathfrak{X}}_x \rightarrow \mathfrak{N}$ satisfying*

$$j \circ (\phi_x)_\eta = f|_{(\widehat{\mathfrak{X}}_x)_\eta}.$$

Such a morphism g is unique.

Theorem B is proved in Section 3. First one extends over a π -adic dilatation of an ambient smooth formal scheme. A section of its reduction, after perfection in positive residue characteristic, makes the reductions of the completed extensions vary continuously. The bounded-function theorem for normal formal schemes then gives formal maps on open neighbourhoods.

To construct $\mathfrak{P}$, we use potential uniformization, ramified descent, and canonical group smoothening. The latter preserves the whole testing category.

Theorem C (Smoothening and construction). *Let $h: \mathfrak{P} \rightarrow \mathfrak{Y}$ be the canonical group smoothening of a flat separated quasi-compact π -adic formal R -group with smooth generic fibre, as in Input 5.1. Then*

$$\text{Map}_R(\mathfrak{X}, \mathfrak{P}) \xrightarrow{\sim} \text{Map}_R(\mathfrak{X}, \mathfrak{Y}) \quad (\mathfrak{X} \in \text{SmForm}_R). \quad (5)$$

If G is smooth, separated, quasi-compact, geometrically connected, and commutative, the uniformization and lifting input of Section 6 yields a quasi-compact model satisfying (3). Its identity completion represents the pointed analytic mapping functor on formal polydiscs. Its Lie algebra is the lattice of derivatives of pointed analytic curves. For an abelian variety A/K , applying this construction to $G = A^{\text{an}}$ and using the gluing theorem in Section 9 gives the finite-type algebraic Néron model of A .

The construction develops the formal-disc argument of [19], within the analytic approach of Bosch–Lütkebohmert [7]. Kappen treats smooth formal schemes of formally finite type using uniformly rigid generic fibres [15]; his Corollary 6.5 gives a local extension criterion for uniformly rigid morphisms. Theorem B concerns maps of the ordinary adic generic fibres. The example in Section 10.2 distinguishes the mapping properties on SmForm_R and $\text{SmForm}_R^{\text{ad}}$.

Section 2 records the formal and analytic foundations. Section 3 proves Theorem B, and Section 4 deduces Theorem A. Sections 5–7 prove the smoothening and construction assertions. Sections 8–10 identify the identity completion, algebraize, and give examples.

2. SMOOTH FORMAL SCHEMES AND THEIR GENERIC FIBRES

2.1. Formal schemes and completion. All schemes and formal schemes are classical and are viewed as sheaves of spaces. Affine rings and ideals below are discrete. We write Mod_B for B -module spectra and CAlg_R for commutative R -algebras in spectra.

A locally noetherian formal R -scheme is *locally formally of finite type* if its reduced scheme is locally of finite type over k . This is also called locally of pseudo-finite type. We use smoothness in the infinitesimal lifting sense, without requiring the structural morphism to be adic [3]. The notation $\text{Spf}_I B$ always includes the I -adic topology; on a π -adic object we abbreviate $\text{Spf}_\pi B$ to $\text{Spf } B$.

For a noetherian R -algebra B and a finitely generated ideal I , put

$$\widehat{B}_I := \varprojlim_{n \geq 0} B/I^{n+1} \quad \text{in } \text{CAlg}_R, \quad (6)$$

with quotient transition maps and the I -adic topology on the limit. In particular,

$$R\langle T_1, \dots, T_m \rangle = R[\widehat{T_1, \dots, T_m}]_{(\pi)}, \quad R[[T_1, \dots, T_m]] = R[\widehat{T_1, \dots, T_m}]_{(\pi, T)}.$$

If $Z \hookrightarrow \mathfrak{Z}_k$ is closed and $\mathfrak{Z} = \text{Spf } A$ is π -adic, the completion $\widehat{\mathfrak{Z}}_Z$ is $\text{Spf}_a \widehat{A}_a$, where a is the inverse image of the ideal of Z in A .

Proposition 2.1 (Local structure). *Every $\mathfrak{X} \in \text{SmForm}_R$ is Zariski-locally of the form*

$$\mathfrak{X} = \widehat{\mathfrak{Z}}_Z, \quad \mathfrak{Z} \in \text{SmForm}_R^{\text{ad}} \text{ affine, } Z \hookrightarrow \mathfrak{Z}_k \text{ closed and reduced.} \quad (7)$$

Conversely every such completion belongs to SmForm_R . These formal schemes are R -flat and normal. Every closed point $x \in \mathfrak{X}_{\text{red}}$ lifts to an R -section, and for any such section there is a pointed isomorphism

$$\widehat{\mathfrak{X}}_x \simeq \text{Spf}_{(\pi, T)} R[[T_1, \dots, T_m]] \quad (8)$$

for some $m \geq 0$. In particular, a pointed object $\mathfrak{X} \in \text{SmForm}_R$ satisfies $\mathfrak{X}_{\text{red}} \simeq \text{Spec } k$ if and only if it is pointed-isomorphic to a formal open polydisc.

Proof. The factorization of a smooth formal morphism as a completion followed by an adic smooth morphism is [3, Theorem 7.13]. Replacing the centre by its reduction does not change the completion. Conversely, completion morphisms are formally étale, so these completions are smooth.

A closed point of Z is k -rational and lifts to an R -section of $\mathfrak{Z}$. Since the ideal of Z then maps into πR , the section factors through $\widehat{\mathfrak{Z}}_Z$. Formal parameters on $\mathfrak{Z}$ centred at this section give (8). Flatness follows from flatness of completion and of $\mathfrak{Z}/R$. The completed local rings are regular; hence the affine rings of $\mathfrak{X}$ are regular and in particular normal. Here every maximal ideal contains the ideal of definition. Finally, an object whose reduction is $\text{Spec } k$ is affine and already complete for its maximal ideal, giving the last assertion. $\square$

In particular, $\mathfrak{X}_{\text{red}}$ is Jacobson.

2.2. The relative adic generic fibre. For a locally noetherian formal R -scheme, set

$$\mathfrak{X}_\eta = \mathfrak{X}^{\text{ad}} \times_{\text{Spa}(R, R)} \text{Spa}(K, R). \quad (9)$$

For $\mathfrak{X} = \text{Spf } B$ flat and topologically of finite type, this is $\text{Spa}(B[1/\pi], B^{\text{int}})$, where B^{int} is the integral closure of B in $B[1/\pi]$. For a formal scheme locally formally of finite type, $\mathfrak{X}_\eta$ is an analytic K -space locally of finite type. Generic fibres, products, and analytifications use Huber's conventions [13, 14]. They extend to algebraic spaces by étale descent. In particular $\mathcal{F}_G$ is a sheaf on the site specified above.

For an affine formal target $\mathfrak{Z} = \text{Spf}_I C$ locally formally of finite type, the generic-fibre mapping property is

$$\text{Map}_K(X, \mathfrak{Z}_\eta) \simeq \text{Hom}_{R\text{-alg}}^{\text{cont}}(C, \mathcal{O}^+(X)). \quad (10)$$

Here X is any analytic K -space, $\mathcal{O}^+(X)$ denotes its functions bounded by one at all adic points, and continuity is checked after restriction to affinoid opens of X , with their analytic topologies. This is the mapping property of $\text{Spa}(C, C)$, followed by the base change in (9); see [23, Proposition 8.24].

Write

$$\mathbb{D}_R^m = \text{Spf}_{(\pi, T)} R[[T_1, \dots, T_m]], \quad D_K^m = (\mathbb{D}_R^m)_\eta.$$

The closed unit polydisc is $\overline{D}_K^m = \text{Spa}(K\langle T \rangle, R\langle T \rangle)$, and

$$D_K^m = \bigcup_{n \geq 1} \{x \in \overline{D}_K^m : |T_i(x)|^n \leq |\pi(x)| \text{ for all } i\}. \quad (11)$$

At rank-one points this domain is described by $|T_i| < 1$. For a closed $x \in \mathfrak{X}_{\text{red}}$, the *residue tube* is

$$]x[_{\mathfrak{X}} := (\widehat{\mathfrak{X}}_x)_\eta \hookrightarrow \mathfrak{X}_\eta.$$

For $\mathfrak{X} \in \text{SmForm}_R$ it is an open polydisc by (8).

Lemma 2.2. *For a separated formal R -scheme $\mathfrak{X}$ locally formally of finite type and a finite extension S/R with fraction field L , generic fibre identifies*

$$\text{Map}_R(\text{Spf } S, \mathfrak{X}) \simeq \text{Map}_K(\text{Spa}(L, S), \mathfrak{X}_\eta).$$

Every such analytic point lies in the tube of a closed point of $\mathfrak{X}_{\text{red}}$. If two maps from an R -flat formal scheme locally formally of finite type to a separated formal target locally formally of finite type agree on generic fibres, they agree formally.

Proof. On $\text{Spf}_I B$, an analytic point is a continuous homomorphism $B \rightarrow L$ taking B into S . It is thus a continuous map to S , with I mapping into its maximal ideal. Reduction gives a k -point and the map factors through the corresponding maximal-ideal completion. These affine descriptions glue; separatedness gives uniqueness. For the final assertion, the ideal of the diagonal pulls back to zero on the generic fibre and hence to zero on the flat source. This faithfulness follows from the affine generic-fibre construction, or from [10, §7.1]. $\square$

2.3. Polydiscs and affine formal targets.

Lemma 2.3 (Bounded functions on a polydisc). *An analytic power series $f = \sum_v a_v T^v$ on D_K^m is bounded by one if and only if every a_v belongs to R . Consequently*

$$\mathcal{O}^+(D_K^m) = R[[T_1, \dots, T_m]].$$

Proof. On a closed subpolydisc of radius $r < 1$ in the divisible value group, the Gauss norm is $\|f\|_r = \max_v |a_v| r^{|v|}$. Boundedness and $r \rightarrow 1$ give $|a_v| \leq 1$. Conversely, an integral power series defines a continuous map $R\langle Z \rangle \rightarrow R[[T]]$. Its generic fibre takes values in the closed unit disc, so it is bounded by one at every adic point. See [6, §2.2] for the Gauss norm. $\square$

Lemma 2.4 (Pointed maps to an affine formal scheme). *Let $\mathfrak{Z} = \mathrm{Spf}_I C$ be an affine noetherian formal R -scheme locally formally of finite type, with a formal section $z: \mathrm{Spf} R \rightarrow \mathfrak{Z}$. Then*

$$\mathrm{Map}_{R,*}(\mathbb{D}_R^m, \mathfrak{Z}) \xrightarrow{\sim} \mathrm{Map}_{K,*}(D_K^m, \mathfrak{Z}_\eta),$$

where pointedness on the two sides refers to z and z_η , respectively.

Proof. By (10) and Lemma 2.3, a pointed analytic map gives

$$\varphi: C \longrightarrow \mathcal{O}^+(D_K^m) = R[[T]], \quad \mathrm{ev}_0 \circ \varphi = z^*.$$

The given formal section satisfies $z^*(I) \subseteq \pi R$. Hence $\varphi(I) \subseteq (\pi, T)$, so φ is continuous for the formal topologies and defines the required formal map. Its generic fibre is the prescribed map by (10). Uniqueness follows from the injection $R[[T]] \hookrightarrow \mathcal{O}(D_K^m)$. $\square$

2.4. Factorization through a tube. For a locally noetherian formal R -scheme $\mathfrak{Y}$ locally formally of finite type and a reduced closed subscheme $Z \hookrightarrow \mathfrak{Y}_{\mathrm{red}}$, put

$$]Z[_{\mathfrak{Y}} := (\widehat{\mathfrak{Y}}_Z)_\eta \hookrightarrow \mathfrak{Y}_\eta.$$

A formal morphism $\mathfrak{X} \rightarrow \mathfrak{Y}$ factors uniquely through $\widehat{\mathfrak{Y}}_Z$ whenever its reduced morphism factors through Z . This is the universal property of completion; see [21, Tag 0GVP]. The completion is affine when Z is affine, by [21, Tag 0DE8].

Lemma 2.5. *Let $f: X \rightarrow \mathfrak{Y}_\eta$ be analytic. Suppose that X has an étale covering $\{(\mathfrak{X}_i)_\eta \rightarrow X\}$, where the $\mathfrak{X}_i$ are locally noetherian formal R -schemes locally formally of finite type, and each restricted map is the generic fibre of $f_i: \mathfrak{X}_i \rightarrow \mathfrak{Y}$. If every $(f_i)_{\mathrm{red}}$ factors through Z , then f factors uniquely through $]Z[_{\mathfrak{Y}}$.*

Proof. Each f_i factors through $\widehat{\mathfrak{Y}}_Z$ by the completion property. The generic-fibre factorizations agree on overlaps because $]Z[_{\mathfrak{Y}} \hookrightarrow \mathfrak{Y}_\eta$ is an open immersion, and étale descent gives the result. $\square$

2.5. Affine targets and detection by points.

Input 2.6 (Bounded functions on a normal formal scheme). If $\mathfrak{X}$ is R -flat, normal, and locally formally of finite type, then restriction gives

$$\Gamma(\mathfrak{X}, \mathcal{O}_{\mathfrak{X}}) \xrightarrow{\sim} \Gamma(\mathfrak{X}_\eta, \mathcal{O}_{\mathfrak{X}_\eta}^+). \quad (12)$$

This is de Jong's theorem [10, Theorem 7.4.1]; see also [16, Corollary 6.3].

Corollary 2.7 (Lifting to an affine formal target). *Let $\mathfrak{X}$ be an R -flat normal formal scheme locally formally of finite type, and let $\mathfrak{U}$ be an affine π -adic formal R -scheme topologically of finite type. Then*

$$\mathrm{Map}_R(\mathfrak{X}, \mathfrak{U}) \xrightarrow{\sim} \mathrm{Map}_K(\mathfrak{X}_\eta, \mathfrak{U}_\eta).$$

Proof. Write $\mathfrak{U} = \mathrm{Spf}_\pi B$. On an affine formal open $\mathrm{Spf}_I A \subset \mathfrak{X}$, the mapping property (10) and (12) associate to an analytic map a homomorphism $B \rightarrow A$. It is continuous because π is topologically nilpotent in A , and its generic fibre is the prescribed map. Lemma 2.2 gives uniqueness and glues these maps over $\mathfrak{X}$. $\square$

Lemma 2.8 (Detection by finite-extension-valued points). *Let $U \hookrightarrow X$ be a quasi-compact open immersion of analytic K -spaces. If $U(L) \rightarrow X(L)$ is surjective for every finite extension L/K , then $U = X$.*

Consequently, if V is separated and $U \subset V$ is a quasi-compact open subspace, a morphism $f: X \rightarrow V$ factors through U if and only if $f(X(L)) \subseteq U(L)$ for every finite extension L/K .

Proof. For an affinoid open $W \subset X$, the open $U \cap W$ is quasi-compact. The affinoid point-detection property gives $U \cap W = W$; see [1, proof of Lemma 2.27]. These equalities prove the first assertion. For the second, $U \rightarrow V$ is quasi-compact: for any affinoid $W \rightarrow V$, the fibre product $W \times_V U$ is closed in the quasi-compact space $W \times_K U$. Apply the first assertion to $X \times_V U \hookrightarrow X$. $\square$

2.6. Schematic closure in a π -adic model. Let $\mathfrak{W}$ be R -flat, π -adic, and locally topologically of finite type. A closed analytic subspace $Z \subset \mathfrak{W}_\eta$ has a flat schematic closure $\overline{Z}_{\mathfrak{W}}^{\text{fl}}$. On $\text{Spf } B \subset \mathfrak{W}$, contract its analytic ideal from $B[1/\pi]$ to an ideal $\mathfrak{b} \subset B$; the closure has ring

$$B/\mathfrak{b} \simeq \varprojlim_n B/(\mathfrak{b} + \pi^{n+1}B).$$

The ideal is π -saturated. On flat formal tests $\mathfrak{X}$ locally formally of finite type, it has the factorization property

$$\text{Map}_R(\mathfrak{X}, \overline{Z}_{\mathfrak{W}}^{\text{fl}}) \simeq \text{Map}_R(\mathfrak{X}, \mathfrak{W}) \times_{\text{Map}_K(\mathfrak{X}_\eta, \mathfrak{W}_\eta)} \text{Map}_K(\mathfrak{X}_\eta, Z). \quad (13)$$

This follows by pulling back the defining ideal and using flatness and generic-fibre faithfulness.

Completion at a closed point is compatible with flat closure and restricts the generic fibre to the corresponding tube. A flat closed formal subscheme of a complete noetherian local formal scheme of formally finite type is determined by its generic fibre. We use these formal-model properties in the completed-graph argument; see [6, Chapters 7–8], [10, §7.1], and [19, §7.2].

3. EXTENSION FROM COMPLETED LOCAL NEIGHBOURHOODS

The proof of Theorem B uses a π -adic dilatation of a smooth ambient model. We first record its construction and the property of its special fibre that will be needed.

3.1. Dilatations. Let $\mathfrak{T}$ be an R -flat π -adic formal scheme locally topologically of finite type, and let $C \hookrightarrow \mathfrak{T}_k$ be closed. On the étale site $\text{FlatForm}_R^{\text{ad}}$ of R -flat π -adic formal schemes locally topologically of finite type, consider the sheaf

$$\mathfrak{X} \longmapsto \text{Map}_R(\mathfrak{X}, \mathfrak{T}) \times_{\text{Map}_k(\mathfrak{X}_k, \mathfrak{T}_k)} \text{Map}_k(\mathfrak{X}_k, C). \quad (14)$$

The formal dilatation $\text{Dil}_C(\mathfrak{T})$ represents this sheaf. The following construction proves representability; compare [18, Proposition 2.6] for the algebraic mapping property.

On an affine chart $\text{Spf } B \subset \mathfrak{T}$, let $\mathfrak{a} \subset B$ be the inverse image of the ideal of $C \cap \text{Spec}(B/\pi B)$. Thus $\pi \in \mathfrak{a}$. Form

$$D_0(B, \mathfrak{a}) := \text{colim}_{m \geq 0} (B \xrightarrow{\pi} \mathfrak{a} \xrightarrow{\pi} \mathfrak{a}^2 \xrightarrow{\pi} \cdots) \quad \text{in } \text{Mod}_B. \quad (15)$$

Here the term of index m is the ordinary ideal power $\mathfrak{a}^m$, regarded as a discrete B -module. Multiplication of ideal powers induces a commutative B -algebra structure on this colimit, with unit from $B = \mathfrak{a}^0$. Define its completion by

$$D(B, \mathfrak{a}) := \varprojlim_{n \geq 0} (D_0(B, \mathfrak{a}) / \pi^{n+1} D_0(B, \mathfrak{a})) \quad \text{in } \text{CAlg}_R, \quad (16)$$

with the quotient transition maps and the π -adic topology on the limit.

The maps $\mathfrak{a}^m \rightarrow B[1/\pi]$, $a \mapsto a/\pi^m$, identify $D_0(B, \mathfrak{a})$ with

$$\bigcup_{m \geq 0} \pi^{-m} \mathfrak{a}^m = B[a/\pi \mid a \in \mathfrak{a}] \subseteq B[1/\pi]. \quad (17)$$

This is the usual affine blowup algebra [21, Tags 052Q, 07Z3]. It is a noetherian R -flat algebra of finite type over B . Its π -adic completion is noetherian, R -flat, and topologically of finite type over R .

Lemma 3.1. *The sheaf (14) is represented over $\mathrm{Spf} B$ by $\mathrm{Spf}_\pi D(B, \mathfrak{a})$. More generally, let A be an R -flat algebra complete for an adic topology in which π is topologically nilpotent. A continuous map $f: B \rightarrow A$ extends to $D(B, \mathfrak{a})$ if and only if $f(\mathfrak{a}) \subseteq \pi A$, and the extension is unique.*

Proof. If $f(\mathfrak{a}) \subseteq \pi A$, then $f(\mathfrak{a}^m) \subseteq \pi^m A$. Division by π^m is unique because A is R -flat. The resulting maps $\mathfrak{a}^m \rightarrow A$ are compatible with the transitions in (15) and with multiplication, so give a unique algebra map $D_0(B, \mathfrak{a}) \rightarrow A$. It is continuous for the π -adic topology on the source. Completeness of A gives its unique extension to (16). Conversely, the relations $\pi(a/\pi) = a$ in (17) force the stated divisibility condition.

For $\mathfrak{X} = \mathrm{Spf}_\pi A$, this condition says exactly that the special-fibre map factors through C . Thus the affine construction represents (14) over $\mathrm{Spf} B$. These representing identifications agree under restriction to affine formal opens. Étale descent therefore glues them to a flat formal scheme $\mathrm{Dil}_C(\mathfrak{T})$, affine over $\mathfrak{T}$. $\square$

If $\mathfrak{T}$ is a formal group and C is a subgroup, (14) is a group object; its representative inherits the group structure by Yoneda.

For $A = R[[T]]$ with its (π, T) -adic topology, the lifting condition is that the induced map $\mathrm{Spec} k[[T]] \rightarrow \mathfrak{T}_k$ factor through C .

For example, take $B = R\langle x \rangle$ and $\mathfrak{a} = (\pi, x)$. Then

$$D(B, \mathfrak{a}) \simeq R\langle u \rangle, \quad x \mapsto \pi u.$$

Its generic fibre is the rational open subspace $|x| \leq |\pi|$ of $\mathfrak{T}_\eta$.

3.2. The special fibre of a dilatation. Let $\mathfrak{T} = \mathrm{Spf} A \in \mathrm{SmForm}_R^{\mathrm{ad}}$ be affine, and let $Z \hookrightarrow \mathfrak{T}_k$ be closed and reduced, with inverse-image ideal $\mathfrak{a} \subset A$. Put

$$\mathfrak{X} = \widehat{\mathfrak{T}}_Z, \quad \mathfrak{Y} = \mathrm{Dil}_Z(\mathfrak{T}).$$

The morphism $\mathfrak{Y} \rightarrow \mathfrak{T}$ factors through a canonical morphism

$$\beta: \mathfrak{Y} \longrightarrow \mathfrak{X}. \tag{18}$$

Indeed, $\mathfrak{a}\mathcal{O}_\mathfrak{Y} \subseteq \pi\mathcal{O}_\mathfrak{Y}$, so the defining homomorphism extends to the $\mathfrak{a}$ -adic completion. Reduction gives an ordinary scheme morphism

$$\beta_0: \mathfrak{Y}_k \longrightarrow Z.$$

The generic fibre of (18) is the open domain in $\mathfrak{X}_\eta$ on which generators of $\mathfrak{a}$ have absolute value at most $|\pi|$.

Lemma 3.2. *There is a universal homeomorphism $q: Z' \rightarrow Z$ and a morphism $\sigma: Z' \rightarrow \mathfrak{Y}_k$ with $\beta_0\sigma = q$. One may take $Z' = Z$ in residue characteristic zero and $Z' = Z^{\mathrm{perf}}$ in positive residue characteristic. In particular, the continuous map $|\mathfrak{Y}_k| \rightarrow |Z|$ admits a section.*

Proof. Put $B = \Gamma(Z', \mathcal{O}_{Z'})$. We first give a flat π -adically complete R -algebra C with $C/\pi C = B$.

In equal characteristic choose a coefficient field, identify $R = k[[\pi]]$, and take $C = B[[\pi]]$. In mixed characteristic choose a Cohen embedding $W(k) \hookrightarrow R$ and take

$$C = R \otimes_{W(k)} W(B).$$

Here B is perfect, so $W(B)$ is p -torsion-free with $W(B)/p = B$. It is flat over the discrete valuation ring $W(k)$. Thus C is R -flat and $C/\pi C = B$. Since R is finite free over $W(k)$, C is complete for p ,

and hence for π . These properties of Witt vectors over perfect rings are standard; compare [4, §3 and Remark 3.15].

The map $A \rightarrow B$ defining $Z' \rightarrow Z \hookrightarrow \mathfrak{T}_k$ lifts, by smoothness of the affine formal scheme $\mathfrak{T}$, through the successive quotients $C/\pi^{n+1}C$. Taking the compatible limit gives a continuous homomorphism $A \rightarrow C$. Its image of $\mathfrak{a}$ lies in πC . Lemma 3.1 therefore gives a unique extension $D(A, \mathfrak{a}) \rightarrow C$. Reduction modulo π defines σ , and its composite with β_0 is q by construction. Since q is a homeomorphism on underlying spaces, $|\sigma| |q|^{-1}$ is a continuous section. $\square$

3.3. A completed graph in a π -adic source. The next lemma is the completed-graph argument of [19, §7.2], applied to a flat π -adic source.

Lemma 3.3. *Let $\mathfrak{Y}$ be an R -flat π -adic formal scheme locally topologically of finite type. Let $\mathfrak{N}$ be an R -flat separated π -adic formal scheme locally topologically of finite type, and let $j: \mathfrak{N}_\eta \hookrightarrow V$ be an open immersion into a separated analytic K -space. Suppose that $f: \mathfrak{Y}_\eta \rightarrow V$ extends to a formal map $\widehat{\mathfrak{Y}}_y \rightarrow \mathfrak{N}$ for every closed $y \in \mathfrak{Y}_k$. Then f extends uniquely to $\mathfrak{Y} \rightarrow \mathfrak{N}$.*

Proof. Work with $\mathfrak{Y}$ affine and fix a closed y . Write $\phi: \widehat{\mathfrak{Y}}_y \rightarrow \mathfrak{N}$ for the prescribed extension and n for the image of its closed point. Choose an affine formal open $\mathfrak{U} \subset \mathfrak{N}$ containing n . The analytic relation

$$Z_\eta = \mathfrak{Y}_\eta \times_V \mathfrak{U}_\eta \hookrightarrow \mathfrak{Y}_\eta \times \mathfrak{U}_\eta$$

is closed because V is separated. Its flat schematic closure $\mathfrak{Z}$ exists in the π -adic model $\mathfrak{Y} \times \mathfrak{U}$.

The formal graph of ϕ defines a map $\widehat{\mathfrak{Y}}_y \rightarrow \mathfrak{Y} \times \mathfrak{U}$ with generic fibre in Z_η . Since $\widehat{\mathfrak{Y}}_y$ is R -flat, (13) factors this map through $\mathfrak{Z}$; in particular, $(y, n) \in \mathfrak{Z}$. On completing at this point, the analytic relation is exactly the graph of ϕ_η , because $\mathfrak{U}_\eta \hookrightarrow V$ is a monomorphism. The closure and completion properties in Section 2 therefore identify $\widehat{\mathfrak{Z}}_{(y,n)}$ with the formal graph of ϕ . The projection induces

$$\widehat{\mathcal{O}}_{\mathfrak{Y},y} \xrightarrow{\sim} \widehat{\mathcal{O}}_{\mathfrak{Z},(y,n)}. \quad (19)$$

Both residue fields are k . The projection is adic and locally of finite presentation, so the completed-local-ring criterion implies that it is étale at (y, n) . Restricting to an étale neighbourhood gives a formal lift of f there.

The resulting étale neighbourhoods cover $\mathfrak{Y}$, since their images contain all closed points of the Jacobson scheme $\mathfrak{Y}_k$. On their fibre products the two lifts have equal generic fibres, because j is a monomorphism, and agree by Lemma 2.2. Étale descent gives the required map and the same lemma gives uniqueness. $\square$

3.4. Proof of the extension theorem.

Proof of Theorem B. Necessity is restriction to completed local neighbourhoods; uniqueness is Lemma 2.2. We prove sufficiency. For every closed $x \in \mathfrak{X}_{\text{red}}$, let $\phi_x: \widehat{\mathfrak{X}}_x \rightarrow \mathfrak{N}$ be its unique extension and let $n_x \in \mathfrak{N}_k(k)$ be the image of the closed point.

(1) An ambient smooth scheme. By Proposition 2.1 and uniqueness, the problem is local on $\mathfrak{X}$. Write $\mathfrak{X} = \widehat{\mathfrak{T}}_Z$ with $\mathfrak{T}$ smooth affine and π -adic, and Z reduced. Form $\mathfrak{Y} = \text{Dil}_Z(\mathfrak{T})$ and $\beta: \mathfrak{Y} \rightarrow \mathfrak{X}$ as in (18).

(2) Extension on the dilatation. For every closed $y \in \mathfrak{Y}_k$, its image $x = \beta_0(y)$ is a closed point of Z . The induced map on completions, followed by ϕ_x , extends $f \circ \beta_\eta$. Lemma 3.3 gives a unique map

$$u: \mathfrak{Y} \longrightarrow \mathfrak{N}, \quad j \circ u_\eta = f \circ \beta_\eta.$$

Uniqueness on $\widehat{\mathfrak{Y}}_y$ implies $u_k(y) = n_x$.

(3) Formal open neighbourhoods. Choose $q: Z' \rightarrow Z$ and $\sigma: Z' \rightarrow \mathfrak{Y}_k$ from Lemma 3.2. The continuous map

$$\rho := |u_k \sigma| \circ |q|^{-1}: |Z| \longrightarrow |\mathfrak{N}_k|$$

satisfies $\rho(x) = n_x$ at every closed point. Indeed, the point of Z' above x is k -rational, so its image under σ is a closed point of $\mathfrak{Y}_k$ to which step (2) applies. For an affine formal open $\mathfrak{U} \subset \mathfrak{N}$, let $\mathfrak{X}_{\mathfrak{U}} \subset \mathfrak{X}$ be the formal open with underlying space $\rho^{-1}(|\mathfrak{U}_k|)$. These opens cover $\mathfrak{X}$. If $x \in \mathfrak{X}_{\mathfrak{U}}$ is closed, then ϕ_x factors through $\mathfrak{U}$.

(4) Formal lifts. Every finite-extension-valued point of $(\mathfrak{X}_{\mathfrak{U}})_{\eta}$ lies in the tube of a closed point of $\mathfrak{X}_{\mathfrak{U}}$, by Lemma 2.2. The preceding factorization of ϕ_x therefore puts its image under f in $\mathfrak{U}_{\eta}$. Lemma 2.8 gives

$$f|_{(\mathfrak{X}_{\mathfrak{U}})_{\eta}}: (\mathfrak{X}_{\mathfrak{U}})_{\eta} \longrightarrow \mathfrak{U}_{\eta}.$$

The source is normal and R -flat, so Corollary 2.7 gives a formal map $\mathfrak{X}_{\mathfrak{U}} \rightarrow \mathfrak{U}$. Their composites to $\mathfrak{N}$ agree on overlaps by Lemma 2.2, and glue to the required morphism $\mathfrak{X} \rightarrow \mathfrak{N}$. $\square$

4. THE POLYDISC CRITERION FOR A FORMAL NÉRON MODEL

Fix $i: \mathfrak{P}_{\eta} \hookrightarrow G$ as in Theorem A, with $\mathfrak{P} \in \text{SmForm}_R^{\text{ad}}$, and assume (2). We use multiplicative notation; commutativity is not required.

4.1. The quotient sheaf. Work in $\mathcal{E}_R = \text{Shv}_{\text{ét}}(\text{SmForm}_R, \text{Spaces})$. Put $J = G(K)$, $H = \mathfrak{P}(R) \subset J$, and $I = J/H$, with quotient map $q: J \rightarrow I$. Define

$$\mathcal{N}_{\mathfrak{P}} := (\underline{J} \times h_{\mathfrak{P}}) // \underline{H}, \quad (g, x) \cdot h = (gh, h^{-1}x). \quad (20)$$

The map $(g, x) \mapsto q(g)$ descends to $\mathcal{N}_{\mathfrak{P}} \rightarrow \underline{I}$. The invariant map $(g, x) \mapsto g i(x)$ induces

$$\mathcal{N}_{\mathfrak{P}} \longrightarrow \mathcal{F}_G. \quad (21)$$

Lemma 4.1. *The sheaf $\mathcal{N}_{\mathfrak{P}}$ is represented on SmForm_R by a separated formal scheme $\mathfrak{N}_{\mathfrak{P}} \in \text{SmForm}_R^{\text{ad}}$. It is quasi-compact exactly when I is finite, and its generic-fibre map $\mathfrak{N}_{\mathfrak{P}, \eta} \rightarrow G$ is an open immersion.*

Proof. The quotient I is discrete because right translation by H is free. For $a \in I$, choose $g \in J$ with $q(g) = a$. The map $x \mapsto (g, x)$ identifies $h_{\mathfrak{P}}$ with the fibre of $\mathcal{N}_{\mathfrak{P}}$ over a . These fibres are therefore representable; their disjoint union represents $\mathcal{N}_{\mathfrak{P}}$ on the whole site. The resulting formal scheme is smooth, separated, and π -adic. Since $\mathfrak{P}$ is nonempty and quasi-compact, its disjoint union of copies is quasi-compact exactly when I is finite.

On the fibre labelled by $q(g)$, the generic-fibre map is $x \mapsto g i(x)$. If the images for g and g' meet over a valued extension, then $g^{-1}g'$ belongs there to $i(\mathfrak{P}_{\eta})$. Membership of this K -point in the open subspace descends, so $g^{-1}g' \in i(\mathfrak{P}_{\eta})(K) = H$ by Lemma 2.2. Thus $q(g) = q(g')$. The images of distinct fibres are disjoint, proving that the map is an open immersion. $\square$

4.2. Proof of the criterion.

Proof of Theorem A. Let $\mathfrak{N} = \mathfrak{N}_{\mathfrak{P}}$ be the formal scheme in Lemma 4.1. Given $\mathfrak{X} \in \text{SmForm}_R$ and $f: \mathfrak{X}_{\eta} \rightarrow G$, consider a closed point $x \in \mathfrak{X}_{\text{red}}$. Choose an R -section through x and regard it as a section t of $\widehat{\mathfrak{X}}_x$. By Proposition 2.1, there is a pointed isomorphism $(\widehat{\mathfrak{X}}_x, t) \simeq (\mathbb{D}_R^m, 0)$ for some m . Put $g = f(t_K)$. The polydisc hypothesis extends $g^{-1}f$ uniquely to a pointed map $v: \widehat{\mathfrak{X}}_x \rightarrow \mathfrak{P}$. The pair (g, v) in (20) gives an extension of f to $\widehat{\mathfrak{X}}_x \rightarrow \mathfrak{N}$.

Thus the hypothesis of Theorem B holds at every closed point of $\mathfrak{X}_{\text{red}}$. It supplies a unique extension $\mathfrak{X} \rightarrow \mathfrak{N}$. Hence (21) is an equivalence on all of SmForm_R , proving (3). Lemma 4.1 supplies the remaining geometric assertions.

The pointwise group structure of $\mathcal{F}_G$ transports to $\mathfrak{N}$ by Yoneda, and its map to G is a homomorphism. $\square$

Corollary 4.2. *The models with property (3) are functorial in analytic homomorphisms and compatible with finite products whenever they exist. Their space of representatives equipped with the representing equivalence is contractible. Their restriction to $\mathbf{SmForm}_R^{\text{ad}}$ is the usual formal Néron model.*

Proof. Apply Yoneda to postcomposition and to $\mathcal{F}_{G \times G'} \simeq \mathcal{F}_G \times \mathcal{F}_{G'}$. The last assertion is restriction of (3). $\square$

5. CANONICAL SMOOTHENING ON SMOOTH FORMAL TESTS

We now prove the smoothening assertion of Theorem C. All formal groups in this section are π -adic and locally topologically of finite type. Dilatations are those constructed in Section 3.

Input 5.1 (Canonical group smoothening). For a flat separated quasi-compact π -adic formal group $\mathfrak{Y}$, locally topologically of finite type and with smooth generic fibre, let $C(\mathfrak{Y}) \subset \mathfrak{Y}_k$ be the reduced closed subgroup generated by the reductions of $\mathfrak{Y}(R)$. Set $\mathfrak{Y}_0 = \mathfrak{Y}$ and $\mathfrak{Y}_{r+1} = \text{Dil}_{C(\mathfrak{Y}_r)}(\mathfrak{Y}_r)$. After finitely many steps this gives a smooth formal group:

$$\mathfrak{P} = \mathfrak{Y}_s \longrightarrow \mathfrak{Y}_{s-1} \longrightarrow \cdots \longrightarrow \mathfrak{Y}_0 = \mathfrak{Y}.$$

The terms remain flat, separated, and quasi-compact; the map $\mathfrak{P}(R) \rightarrow \mathfrak{Y}(R)$ is bijective and $\mathfrak{P}_\eta \rightarrow \mathfrak{Y}_\eta$ is an open immersion.

We use the formal version stated in [19, Input 1.6], based on [8, §7.1, Theorem 5]; see also [22, §2.3.2] for the canonical centres and [12, p. 128] for formal group smoothening. Finite termination is the technical input. At a smooth stage, every closed special-fibre point lifts to an R -section, so $C(\mathfrak{Y}_s) = (\mathfrak{Y}_s)_k$ and the next dilatation is the identity. Thus the full inverse system is eventually constant, and

$$h_{\mathfrak{P}} \simeq \varprojlim_{r \geq 0} h_{\mathfrak{Y}_r} \quad \text{in } \text{Shv}_{\text{ét}}(\mathbf{SmForm}_R, \text{Spaces}). \quad (22)$$

5.1. Stabilization of images of Greenberg groups. Put $R_n = R/\pi^{n+1}$ and $\mathfrak{Y}_n = \mathfrak{Y} \times_R \text{Spec } R_n$. Let $\mathcal{R}_n$ be the Greenberg ring scheme and

$$G_n = \text{Gr}_n(\mathfrak{Y}), \quad G_n(B) = \mathfrak{Y}_n(\mathcal{R}_n(B))$$

for a k -algebra B . These are finite-type algebraic groups over k , with $G_0 = \mathfrak{Y}_k$ and $G_n(k) = \mathfrak{Y}(R_n)$. The construction is functorial in k -algebras; see [4, §§3, 7, 11, 17].

Lemma 5.2. *The reduced scheme-theoretic images*

$$C_n = (\text{im}(G_n \longrightarrow \mathfrak{Y}_k))_{\text{red}}$$

stabilize to $C(\mathfrak{Y})$. More precisely, their stable value C satisfies

$$C(k) = \text{im}(\mathfrak{Y}(R) \longrightarrow \mathfrak{Y}_k(k)). \quad (23)$$

Proof. Images of homomorphisms of algebraic groups are closed. Their k -points are exactly the images of k -points: a nonempty finite-type fibre over the algebraically closed field k has a k -point. The C_n form a descending chain. Noetherianity makes their underlying closed subsets, hence their reduced closed structures, stabilize.

To obtain compatible lifts, apply the same stabilization at every level. For fixed n , let H_n be the stable reduced image of $G_m \rightarrow G_n$ for $m \geq n$. The maps $H_{n+1}(k) \rightarrow H_n(k)$ are surjective. To lift a point of $H_n(k)$, choose m after stabilization at both levels and lift it to $G_m(k)$; its image at level $n+1$ belongs to $H_{n+1}(k)$.

Starting with $c \in H_0(k) = C(k)$, choose compatible lifts in all $H_n(k)$. Completeness gives

$$\mathfrak{Y}(R) = \varprojlim_n \mathfrak{Y}(R_n).$$

For a nonaffine $\mathfrak{Y}$, this equality is checked in an affine open containing the common reduction of the sections. Thus c lifts to an R -point. The reverse inclusion in (23) is immediate. Since a reduced algebraic group over the perfect field k is smooth, and its k -points are dense, C is exactly the reduced closed subgroup generated by those reductions. $\square$

5.2. A lifting criterion.

Lemma 5.3. *Let A be flat over R , complete for an adic topology in which π is topologically nilpotent, and suppose $B = A/\pi A$ is reduced. Assume that for every n there is an R -algebra map*

$$A \longrightarrow \mathcal{R}_n(B) \tag{24}$$

whose reduction to B is the quotient map. Every formal map $\mathrm{Spf} A \rightarrow \mathfrak{Y}$ whose image lies in an affine formal chart then lifts uniquely to $\mathfrak{F}$.

Proof. Composing the defining map of the chart into A with (24) gives a B -point of G_n . Its image in $G_0(B)$ is the same reduction $\mathrm{Spec} B \rightarrow \mathfrak{Y}_k$ for every n . This reduction factors through the scheme-theoretic image of $G_n \rightarrow G_0$, and, since B is reduced, through its reduction C_n . For n after stabilization it factors through $C(\mathfrak{Y})$. Lemma 3.1 gives the unique lift through the first dilatation. The inverse image of the chosen chart under each dilatation is affine. We may therefore repeat the argument with the same source algebra through the finite smoothening sequence. $\square$

5.3. The Witt-vector construction.

Lemma 5.4. *For $A = R[[T_1, \dots, T_m]]$ and $B = k[[T_1, \dots, T_m]]$, maps (24) exist for every n .*

Proof. In mixed characteristic $(0, p)$, choose a Cohen embedding $W(k) \hookrightarrow R$. The extension $R/W(k)$ is finite free. Set $A_0 = W(k)[[T_1, \dots, T_m]]$. Witt Frobenius on coefficients and $T_i \mapsto T_i^p$ define a Frobenius lift σ on A_0 . Since A_0 is p -torsion free,

$$\delta(a) = \frac{\sigma(a) - a^p}{p}$$

defines a δ -structure.

The Witt-vector functor is right adjoint to the forgetful functor from δ -rings to rings [5, Remark 2.4]. Applying the adjunction to reduction $A_0 \rightarrow B$ gives a homomorphism

$$A_0 \longrightarrow W(B)$$

with prescribed zeroth Witt component. It is compatible with $W(k)$ by naturality. Finite freeness gives the tensor-product identification

$$R[[T]] = R \otimes_{W(k)} W(k)[[T]],$$

so we obtain $A \rightarrow R \otimes_{W(k)} W(B)$. Choose a with $p^a R_n = 0$ and compose with truncation and the canonical comparison

$$R_n \otimes_{W_a(k)} W_a(B) \longrightarrow \mathcal{R}_n(B).$$

This comparison exists for every k -algebra B by [4, Proposition 3.14]. All maps commute with reduction. This proves (24).

In equal characteristic, a coefficient field identifies R with $k[[\pi]]$ and $\mathcal{R}_n(B)$ with $B[\pi]/\pi^{n+1}$. Truncation of the π -expansion gives the required map. $\square$

5.4. The mapping property of smoothening.

Theorem 5.5. *For the canonical smoothening $h: \mathfrak{P} \rightarrow \mathfrak{Y}$ of Input 5.1, composition induces*

$$\mathrm{Map}_R(\mathfrak{X}, \mathfrak{P}) \xrightarrow{\sim} \mathrm{Map}_R(\mathfrak{X}, \mathfrak{Y}) \quad (\mathfrak{X} \in \mathrm{SmForm}_R).$$

The equivalence respects pointed maps, composition in the source, and the group law in the target.

Proof. First suppose $\mathfrak{X}_{\mathrm{red}} = \mathrm{Spec} k$. By Proposition 2.1, the source admits a presentation $\mathrm{Spf}_{(\pi, T)} R[[T]]$. Every map to $\mathfrak{Y}$ factors through an affine formal chart. Lemmas 5.3 and 5.4 therefore lift it uniquely through the smoothening.

Now let $\mathfrak{X} = \mathrm{Spf}_I A \in \mathrm{SmForm}_R$ be affine and let $f: \mathfrak{X} \rightarrow \mathfrak{Y}_i$ be a map into one stage. Work in an affine chart of the target. For each maximal ideal $\mathfrak{m} \subset A$, the source $\mathrm{Spf}_{\mathfrak{m}} \widehat{A}_{\mathfrak{m}}$ has reduced scheme $\mathrm{Spec} k$. The first part gives a lift through the next dilatation. Thus the inverse-image ideal of $C(\mathfrak{Y}_i)$ maps into $\pi \widehat{A}_{\mathfrak{m}}$ for every $\mathfrak{m}$.

Since I lies in the Jacobson radical of A , these maximal ideals are exactly the closed points of $\mathfrak{X}_{\mathrm{red}}$. Faithful flatness of the completed local rings gives an injection

$$A/\pi A \longrightarrow \prod_{\mathfrak{m} \in \mathrm{Max} A} \widehat{A}_{\mathfrak{m}}/\pi \widehat{A}_{\mathfrak{m}}.$$

Consequently the centre ideal maps into πA . Lemma 3.1 gives a unique lift through the dilatation. Iterating the finite sequence proves the assertion on these affine opens. Uniqueness glues the lifts and proves all compatibilities. $\square$

This proves (5). It also characterizes $\mathfrak{P} \rightarrow \mathfrak{Y}$ as the representative of $h_{\mathfrak{Y}}|_{\mathrm{SmForm}_R}$, so canonical smoothening is functorial.

6. BOUNDEDNESS AFTER A FINITE EXTENSION

In this section G is smooth, separated, quasi-compact, geometrically connected, and commutative. The formal groups used to construct the integral neighbourhood are π -adic and locally topologically of finite type. We first prove the polydisc extension property for the generic fibre of such a group.

6.1. Analytic polydiscs in a formal group.

Lemma 6.1. *Every pointed analytic map $D_K^m \rightarrow \mathbb{G}_{m,K}^{\mathrm{an}}$ factors through*

$$\{|z| = 1\} = \mathrm{Spa}(K\langle z, z^{-1} \rangle, R\langle z, z^{-1} \rangle) = (\mathrm{Spf}_{\pi} R\langle z, z^{-1} \rangle)_{\eta}$$

and comes from a unique pointed formal map $\mathbb{D}_R^m \rightarrow \mathrm{Spf}_{\pi} R\langle z, z^{-1} \rangle$. The assertion holds for any split torus, coordinatewise.

Proof. On a closed polydisc, the Tate-algebra unit criterion says that the constant coefficient of a unit strictly dominates its nonconstant coefficients. A unit taking the value 1 at the origin therefore has absolute value 1 everywhere. Restricting to the closed subpolydiscs in (11), and making finite extensions to realize their radii, gives the asserted factorization on an étale cover. It descends to D_K^m . Lemma 2.4, with formal basepoint $z = 1$, gives the formal map. $\square$

Input 6.2 (Weil extension). If $\mathfrak{Q}$ is a smooth separated quasi-compact π -adic formal group locally topologically of finite type over a complete DVR, then

$$\mathrm{Map}(\mathfrak{T}, \mathfrak{Q}) \simeq \mathrm{Map}(\mathfrak{T}_{\eta}, \mathfrak{Q}_{\eta})$$

for every smooth π -adic formal test $\mathfrak{T}$ locally topologically of finite type. The assertion remains valid after a finite extension of the base.

This is the identity-immersion case of the Bosch–Schl ter criterion recalled in [15, Theorem 1.2(i)].

Proposition 6.3. *For such a formal group $\mathfrak{Q}/R$ and every $m \geq 0$, generic fibre induces*

$$\mathrm{Map}_{R,*}(\mathbb{D}_R^m, \mathfrak{Q}) \xrightarrow{\sim} \mathrm{Map}_{K,*}(D_K^m, \mathfrak{Q}_\eta).$$

The formal maps factor through $\widehat{\mathfrak{Q}}_{e_k}$.

Proof. The case $m = 0$ is immediate. Chevalley’s theorem applied to the algebraic group $\mathfrak{Q}_k^0$ gives

$$1 \longrightarrow H \longrightarrow \mathfrak{Q}_k^0 \longrightarrow A \longrightarrow 1,$$

with H smooth connected affine and A an abelian variety [9, Theorem 1.1]. Every pointed morphism $\mathbf{A}_k^m \rightarrow \mathfrak{Q}_k$ factors through H : its image is in $\mathfrak{Q}_k^0$, and its composite to A is constant, since abelian varieties contain no rational curves. Restricting to coordinate lines and using reducedness gives this assertion in every dimension. Thus $\widehat{\mathfrak{Q}}_H$ is an affine formal scheme, with the identity section as basepoint.

Let $f: D_K^m \rightarrow \mathfrak{Q}_\eta$ be pointed. On each smaller closed polydisc, after a finite extension realizing its radius, Input 6.2 gives a pointed formal map from a smooth π -adic model with reduction $\mathbf{A}_k^m$. Its reduced map factors through H . These models, regarded as formal R -schemes, have generic fibres forming an  tale cover of D_K^m . Lemma 2.5 yields

$$D_K^m \longrightarrow (\widehat{\mathfrak{Q}}_H)_\eta \hookrightarrow \mathfrak{Q}_\eta.$$

Lemma 2.4 gives a pointed formal lift into $\widehat{\mathfrak{Q}}_H$, hence into $\mathfrak{Q}$. Its reduced image is e_k , so it factors through $\widehat{\mathfrak{Q}}_{e_k}$. Uniqueness is Lemma 2.2. $\square$

6.2. The uniformization input.

Input 6.4 (Uniformization and polydisc lifting). After a finite extension L/K , with valuation ring S , there are a smooth connected quasi-compact π -adic formal group $\mathfrak{Q}/S$, a split formal torus $\mathfrak{T} \subset \mathfrak{Q}$, and a smooth π -adic formal quotient $\mathfrak{B} = \mathfrak{Q}/\mathfrak{T}$. The pushout along $\mathfrak{T}_\eta \subset T^{\mathrm{an}}$ gives

$$0 \longrightarrow T^{\mathrm{an}} \longrightarrow E \longrightarrow \mathfrak{B}_\eta \longrightarrow 0, \quad 0 \longrightarrow M \longrightarrow E \xrightarrow{q} G_L \longrightarrow 0,$$

where M is a constant finitely generated free abelian group and q is  tale. The induced homomorphism $\mathfrak{Q}_\eta \rightarrow G_L$ is an open immersion. For every $m \geq 0$,

$$\mathrm{Map}_{L,*}(D_L^m, E) \xrightarrow{\sim} \mathrm{Map}_{L,*}(D_L^m, G_L). \quad (25)$$

The structure data are supplied by potential uniformization [17, Theorem 8.2 and Corollary 8.3]. We use the package in the form of [19, Input 1.5], together with the polydisc-lifting clause (25) from the lattice-covering argument in [19, proof of Proposition 4.1]. The construction below is relative to this input. The residue field of S is k .

Lemma 6.5. *For every $m \geq 0$, the natural map*

$$\mathrm{Map}_{S,*}(\mathbb{D}_S^m, \mathfrak{Q}) \longrightarrow \mathrm{Map}_{S,*}(\mathbb{D}_S^m, \mathfrak{B}) \times_{\mathrm{Map}_{L,*}(D_L^m, \mathfrak{B}_\eta)} \mathrm{Map}_{L,*}(D_L^m, E) \quad (26)$$

is an equivalence.

Proof. Fix $b: \mathbb{D}_S^m \rightarrow \mathfrak{B}$ pointed. Its pointed formal and analytic lifting spaces are torsors under $\mathrm{Map}_{S,*}(\mathbb{D}_S^m, \mathfrak{T})$ and $\mathrm{Map}_{L,*}(D_L^m, T^{\mathrm{an}})$, respectively. The formal lifting space is nonempty: $S[[T_1, \dots, T_m]]$ is local, so the line bundles defining a split-torus torsor are trivial; a lift can be translated to take the identity value at the origin. Lemma 6.1 identifies the acting groups. The equivariant generic-fibre map between their torsors is therefore an equivalence. $\square$

Proposition 6.6. *For the fixed pair $(S, \mathfrak{Q})$ of Input 6.4, every pointed analytic map $D_L^m \rightarrow G_L$ extends uniquely to $\mathbb{D}_S^m \rightarrow \mathfrak{Q}$, for every $m \geq 0$.*

Proof. Lift to E by (25). The projection to $\mathfrak{B}_\eta$ extends formally by Proposition 6.3. Lemma 6.5 then extends the prescribed analytic lift into $\mathfrak{Q}$. $\square$

6.3. Finiteness of the quotient on rational points. Let $X^*(T)$ be the character lattice and $N_T = \text{Hom}(X^*(T), \mathbb{Z})$ its dual. Choose a uniformizer π_L of S and normalize $v_L(\pi_L) = 1$. Then

$$T(L)/\mathfrak{T}(S) \simeq N_T.$$

We first compute the corresponding quotient of $G(L)$.

Lemma 6.7. *Under Input 6.4, the group $G(L)/\mathfrak{Q}(S)$ is finite.*

Proof. The fibre of $q: E \rightarrow G_L$ over an L -point is a torsor under the constant group M . It is trivial: a continuous homomorphism from the profinite absolute Galois group of L to M has finite, hence zero, image. Thus $E(L) \rightarrow G(L)$ is surjective.

Every point of $\mathfrak{B}_\eta(L)$ extends to $b \in \mathfrak{B}(S)$ by Lemma 2.2. The fibre of $\mathfrak{Q} \rightarrow \mathfrak{B}$ over b is a split torus torsor over the local ring S , so is trivial. Hence b lifts to $\mathfrak{Q}(S)$, and

$$E(L) = T(L) + \mathfrak{Q}(S), \quad T(L) \cap \mathfrak{Q}(S) = \mathfrak{T}(S).$$

The first identity defines a homomorphism $v: E(L) \rightarrow N_T$ with kernel $\mathfrak{Q}(S)$; it restricts to the usual valuation on $T(L)$. Consequently

$$G(L)/\mathfrak{Q}(S) \simeq N_T/v(M). \quad (27)$$

To prove finiteness, choose split torus coordinates $z_1, \dots, z_r$. For $a \in \mathbb{Z}^r$, define a domain $E_a \subset E$ by

$$|\pi_L|^{a_i+1} \leq |z_i| \leq |\pi_L|^{a_i-1} \quad (1 \leq i \leq r) \quad (28)$$

in local formal trivializations of $\mathfrak{Q} \rightarrow \mathfrak{B}$. Changes of trivialization multiply each coordinate by a formal unit, whose absolute value is one at every point of the generic fibre. The rational domains therefore glue. On L -points, their coordinate valuations agree with v . They cover E : at an analytic point, the value of an invertible analytic coordinate is bounded above and below by powers of the topologically nilpotent unit π_L .

The images $q(E_a)$ are open and cover G_L , because q is étale and surjective. Quasi-compactness supplies a finite subcover $q(E_{a_1}), \dots, q(E_{a_s})$. Let $g \in G(L)$ and choose a lift $e \in E(L)$. If $g \in q(E_{a_j})$, a point above g in E_{a_j} differs from e by an element of the constant lattice M . Hence the class of $v(e)$ modulo $v(M)$ has a representative in

$$\prod_{i=1}^r [a_{j,i} - 1, a_{j,i} + 1] \cap \mathbb{Z}^r.$$

The union of these finite sets over j maps onto (27). This proves finiteness. $\square$

7. DESCENT AND SCHEMATIC CLOSURE

Let S/R be a finite extension of complete DVRs, with fraction fields L/K and common residue field k . The extension L/K is separable, and S/R is totally ramified. Suppose that a smooth separated quasi-compact π -adic formal group $\mathfrak{Q}/S$, locally topologically of finite type, is given with an open immersion of groups $\mathfrak{Q}_\eta \hookrightarrow G_L$.

7.1. Formal Weil restriction. For a formal R -scheme $\mathfrak{X} = \mathrm{Spf}_I A$ locally formally of finite type, its base change to S has ring

$$A \widehat{\otimes}_R S := \varprojlim_{n \geq 0} (A/I^{n+1}) \otimes_R S \simeq A \otimes_R S, \quad (29)$$

with the I -adic topology. Finite freeness of S/R gives the last equality. Write $\mathfrak{X}_S$ for this formal base change. It takes SmForm_R to SmForm_S and gives $(\mathbb{D}_R^m)_S \simeq \mathbb{D}_S^m$ with the specified topologies.

Lemma 7.1. *There is a smooth separated quasi-compact π -adic formal group $\mathfrak{W} = \mathrm{Res}_{S/R} \mathfrak{Q}$ with a natural adjunction*

$$\mathrm{Map}_R(\mathfrak{X}, \mathfrak{W}) \simeq \mathrm{Map}_S(\mathfrak{X}_S, \mathfrak{Q}) \quad (30)$$

for every formal R -scheme $\mathfrak{X}$ locally formally of finite type. There is a natural map of analytic étale sheaves

$$\mathfrak{W}_\eta \longrightarrow \mathrm{Res}_{L/K} G_L \quad (31)$$

representable by open immersions.

Proof. For an affine chart $\mathrm{Spf} B$ of $\mathfrak{Q}$, choose an R -basis $s_1, \dots, s_a$ of S . In a presentation

$$B = S\langle x_1, \dots, x_b \rangle / (f_1, \dots, f_c),$$

substitute $x_i = \sum_j s_j x_{ij}$. The coefficient equations for the f_ℓ define a topologically finite-type formal R -scheme with the required adjunction.

For gluing, each map $\mathrm{Spec}(S/\pi^{n+1}S) \rightarrow \mathrm{Spec}(R/\pi^{n+1})$ is a finite universal homeomorphism, and remains so after arbitrary base change. Hence the condition that a map to $\mathfrak{Q}$ land in one affine formal open is an open condition on the original test. The affine restrictions glue and form a finite open cover of $\mathfrak{W}$. This is the totally ramified case of the formal restriction construction; compare [4, Corollary 2.48 and Remark 5.14]. The adjunction transports the group structure from $\mathfrak{Q}$ to $\mathfrak{W}$. Smoothness follows by infinitesimal lifting for a finite flat base change, and separatedness follows from the diagonal. For $\mathfrak{X} = \mathrm{Spf}_I A$, the same substitutions use the complete algebra (29). A map from a π -adic chart ring to $A \otimes_R S$ is continuous because π is topologically nilpotent. The map $\mathfrak{X}_S \rightarrow \mathfrak{X}$ is a homeomorphism on underlying spaces, so the affine adjunction glues for this source topology as well. This proves (30) on the stated full category.

On analytic charts, the coefficient functions x_{ij} are uniquely determined by an analytic map after base change to L . The image of the formal restriction imposes the rational bounds $|x_{ij}| \leq 1$, in addition to the equations. Requiring the map to land in the chosen formal chart's generic fibre is also an open condition. These rational domains glue to prove (31); see [11, §3] and [19, Lemma 5.2]. $\square$

7.2. The descent condition. The analytic restriction sheaf is defined by

$$(\mathrm{Res}_{L/K} G_L)(V) = \mathrm{Map}_L(V_L, G_L).$$

There is a unit map $\delta: G \rightarrow \mathrm{Res}_{L/K} G_L$.

Lemma 7.2. *If G is separated, the map δ is representable by closed immersions.*

Proof. After base change to a finite normal extension splitting L/K , the restriction sheaf becomes a finite product of copies of the base change of G . The unit becomes the diagonal. Separatedness makes it a closed immersion, and representable closed immersions descend along this finite étale cover. $\square$

Define U by the Cartesian square of analytic sheaves

$$\begin{array}{ccc} U & \hookrightarrow & \mathfrak{B}_\eta \\ \downarrow & & \downarrow \\ G & \xhookrightarrow{\delta} & \mathrm{Res}_{L/K} G_L. \end{array} \quad (32)$$

The vertical arrows are open immersions and the horizontal arrows are closed immersions. Thus U is represented, is closed in the quasi-compact space $\mathfrak{B}_\eta$, and is a smooth open subgroup of G .

Definition 7.3. The descended formal group associated to $(S, \mathfrak{Q}, \mathfrak{Q}_\eta \hookrightarrow G_L)$ is

$$\mathfrak{Y} = \overline{U}_{\mathfrak{B}}^{\mathrm{fl}}. \quad (33)$$

Proposition 7.4. *The formal scheme $\mathfrak{Y}$ is flat, separated, and quasi-compact, with a natural group structure and an open immersion $\mathfrak{Y}_\eta = U \hookrightarrow G$. Suppose that every pointed analytic map $D_L^m \rightarrow G_L$ extends uniquely to $\mathbb{D}_S^m \rightarrow \mathfrak{Q}$, for all $m \geq 0$. Then every pointed analytic map $D_K^m \rightarrow G$ extends uniquely to $\mathbb{D}_R^m \rightarrow \mathfrak{Y}$, for all $m \geq 0$. On rational points,*

$$\mathfrak{Y}(R) = G(K) \cap \mathfrak{Q}(S) \quad \text{inside } G(L). \quad (34)$$

Proof. Flat schematic closure gives the first three geometric properties and identifies the generic fibre with U . The product $\mathfrak{Y} \times_R \mathfrak{Y}$ is flat. Its multiplication map to $\mathfrak{B}$ has generic fibre in the subgroup U , so it factors uniquely through $\mathfrak{Y}$. The same argument applies to identity and inversion. The group identities follow from uniqueness of factorization.

For a pointed map $f: D_K^m \rightarrow G$, extend its base change uniquely to $\mathbb{D}_S^m \rightarrow \mathfrak{Q}$. The adjunction (30) gives $\mathbb{D}_R^m \rightarrow \mathfrak{B}$. Its generic fibre factors through the fibre product (32): the two composites to $\mathrm{Res}_{L/K} G_L$ agree by construction. The flat-closure property therefore factors it uniquely through $\mathfrak{Y}$. Its value at the origin is the identity by uniqueness. The open immersion into G and Lemma 2.2 prove uniqueness of the extension.

Finally, $\mathfrak{B}(R) = \mathfrak{Q}(S)$. An R -point of $\mathfrak{B}$ factors through $\mathfrak{Y}$ exactly when its generic fibre factors through U , which by (32) is exactly the condition that its value in $G(L)$ descend to $G(K)$. This proves (34). $\square$

Remark 7.5. If L/K is Galois and $\mathfrak{Q}$ has a compatible semilinear action of $\Gamma = \mathrm{Gal}(L/K)$, then

$$\mathfrak{Y} \simeq ((\mathrm{Res}_{S/R} \mathfrak{Q})^\Gamma)^\Gamma.$$

Here the superscript fl denotes the flat closure of the generic fibre. Indeed, δ identifies G with $(\mathrm{Res}_{L/K} G_L)^\Gamma$. The equivariant monomorphism (31) identifies its pullback with $(\mathfrak{B}_\eta)^\Gamma$, and finite closed equalizers commute with generic fibre. Taking flat closure proves the assertion and recovers the construction of [19, §5].

7.3. Construction of the smooth formal group.

Proposition 7.6. *Take $(S, \mathfrak{Q})$ from Input 6.4, form $\mathfrak{Y}$ by (33), and let $\mathfrak{P} \rightarrow \mathfrak{Y}$ be its canonical smoothening. Then $\mathfrak{P} \in \mathrm{SmForm}_R^{\mathrm{ad}}$ satisfies the polydisc hypothesis (2) of Theorem A, and $G(K)/\mathfrak{P}(R)$ is finite.*

Proof. Propositions 6.6 and 7.4 give the polydisc extension property for $\mathfrak{Y}$. Theorem 5.5 preserves it on passage to $\mathfrak{P}$. Smoothening supplies a smooth separated quasi-compact π -adic formal group and an open immersion $\mathfrak{P}_\eta \hookrightarrow \mathfrak{Y}_\eta \hookrightarrow G$. Moreover,

$$\mathfrak{P}(R) = \mathfrak{Y}(R) = G(K) \cap \mathfrak{Q}(S).$$

Hence $G(K)/\mathfrak{P}(R)$ injects into $G(L)/\mathfrak{Q}(S)$, which is finite by Lemma 6.7. $\square$

Combining Proposition 7.6 with Theorem A proves the existence and quasi-compactness assertions of Theorem C. The resulting $\mathfrak{N}$ and its map to G are independent of the auxiliary extension and the formal model, by Corollary 4.2.

8. THE COMPLETION AT THE IDENTITY AND ITS CARTIER MODULE

Let G be commutative, and let $\mathfrak{P}$ and $\mathfrak{N}$ be a chart and its formal model as in Theorem A. In (20), the map $x \mapsto (e, x)$ induces an open immersion $\mathfrak{P} \hookrightarrow \mathfrak{N}$ through the identity. Consequently

$$\mathfrak{H} := \widehat{\mathfrak{P}}_{e_k} \simeq \widehat{\mathfrak{N}}_{e_k}, \quad \text{Lie}_R(\mathfrak{P}) = \text{Lie}_R(\mathfrak{N}). \quad (35)$$

A pointed map $\mathbb{D}_R^m \rightarrow \mathfrak{N}$ factors through $\mathfrak{H}$. The mapping property on SmForm_R therefore gives

$$\text{Map}_{R,*}(\mathbb{D}_R^m, \mathfrak{H}) \simeq \text{Map}_{K,*}(D_K^m, G) \quad (m \geq 0). \quad (36)$$

The underlying pointed formal scheme of $\mathfrak{H}$ is isomorphic to a formal polydisc. Thus (36), with pointed formal substitutions, characterizes $\mathfrak{H}$ by Yoneda; its group law corresponds to pointwise addition.

8.1. The tangent lattice. Put $\mathfrak{g} = \text{Lie}_K(G)$ and define

$$\Lambda_{\text{disc}} = \{d_0\gamma(1) : \gamma : D_K^1 \rightarrow G, \gamma(0) = e\} \subset \mathfrak{g}.$$

Addition of curves and substitutions $T \mapsto aT$, $a \in R$, make this an R -submodule. For affine $\mathfrak{T} \in \text{SmForm}_R$, sections $t \in \mathfrak{T}(R)$, and analytic maps $f : \mathfrak{T}_\eta \rightarrow G$ with $f(t_K) = e$, let Λ_{sm} be the sum of the images of

$$d_t f : T_t(\mathfrak{T}/R) \longrightarrow \mathfrak{g}.$$

Here $T_t(\mathfrak{T}/R)$ is the dual of the finite free module obtained by pulling the completed relative differentials back along t .

Proposition 8.1. *There are equalities of full R -lattices in $\mathfrak{g}$:*

$$\Lambda_{\text{disc}} = \Lambda_{\text{sm}} = \text{Lie}_R(\mathfrak{N}). \quad (37)$$

Proof. By (36), analytic curves in the definition of Λ_{disc} are exactly formal curves in $\mathfrak{H}$, so their derivatives lie in $\text{Lie}_R(\mathfrak{H})$. Conversely, in formal parameters $X_1, \dots, X_d$, the curve $X_i \mapsto a_i T$ realizes any vector $(a_1, \dots, a_d) \in R^d$. Thus $\Lambda_{\text{disc}} = \text{Lie}_R(\mathfrak{N})$.

Every f in the definition of Λ_{sm} extends to $\mathfrak{N}$ by its mapping property. Its differential on the integral tangent module therefore takes values in $\text{Lie}_R(\mathfrak{N})$. An affine formal neighbourhood of the identity in $\mathfrak{N}$, with its own generic-fibre map, gives the reverse inclusion. Finally, smoothness makes the Lie module finite free, and the generic-fibre open immersion identifies its scalar extension with $\mathfrak{g}$. $\square$

8.2. The Cartier module. Assume the residue characteristic is $p > 0$. The analytic curve group

$$C_G = \text{Map}_{K,*}(D_K^1, G)$$

is canonically identified by (36) with the formal curve group of $\mathfrak{H}$. Transporting the standard p -typical projector and Cartier operators gives the covariant p -typical Cartier module M_G .

We use the following consequence of Cartier theory: the Cartier functor on smooth commutative formal groups is fully faithful, and its tangent quotient is $M(\mathfrak{H})/VM(\mathfrak{H}) \simeq \text{Lie}_R(\mathfrak{H})$. See [24, Chapter IV]. It follows that

$$M_G \simeq M(\mathfrak{H}), \quad M_G/VM_G \simeq \Lambda_{\text{disc}}. \quad (38)$$

Thus M_G determines the identity completion $\mathfrak{H}$.

The normalized formal logarithm over K writes a p -typical curve as

$$\log(c(T)) = \sum_{r \geq 0} b_r T^{p^r}, \quad b_r \in \mathfrak{g}.$$

In the covariant convention,

$$\begin{aligned} \log(Vc)(T) &= \sum_{r \geq 0} b_r T^{p^{r+1}}, \\ \log(Fc)(T) &= p \sum_{r \geq 0} b_{r+1} T^{p^r}, \\ \log([a]c)(T) &= \sum_{r \geq 0} a^{p^r} b_r T^{p^r} \quad (a \in R). \end{aligned}$$

The coefficient b_0 gives the tangent quotient in (38). In residue characteristic zero, R is a $\mathbf{Q}$ -algebra and the normalized formal logarithm identifies $\mathfrak{H}$ directly with the additive formal group of $\mathrm{Lie}_R(\mathfrak{N})$.

9. ALGEBRAIC NÉRON MODELS BY GLUING

Let A/K be a smooth separated algebraic group of finite type. Suppose that $\mathfrak{N}$ is a π -adic formal Néron model of A^{an} with an open immersion $j: \mathfrak{N}_\eta \hookrightarrow A^{\mathrm{an}}$.

Input 9.1 (Gluing and representability by schemes). For the complete DVR R , the functor

$$X \longmapsto (X_K, \widehat{X}, j_X)$$

is an equivalence from separated algebraic spaces locally of finite type over R to triples $(U, \mathfrak{X}, j)$, where U/K is a separated algebraic space locally of finite type, $\mathfrak{X}/R$ is a separated π -adic formal algebraic space locally topologically of finite type, and $j: \mathfrak{X}_\eta \hookrightarrow U^{\mathrm{an}}$ is an open immersion. It preserves products; smoothness and quasi-compactness of a morphism are detected on the algebraic and formal parts. A separated group algebraic space locally of finite type over a noetherian normal one-dimensional base is a scheme.

The equivalence and detection statements are [2, Theorem 1.3 and Proposition 2.27], applicable since a complete DVR is excellent. The last assertion is Raynaud's theorem [20, Theorem 3.3.1]. Its use in this application is also explained in [2, Remark 3.15].

Theorem 9.2. *The triple $(A, \mathfrak{N}, j)$ glues to the algebraic Néron model of A , locally of finite type over R . If $\mathfrak{N}$ is quasi-compact, this model is of finite type.*

Proof. Gluing gives a separated algebraic space $\mathcal{A}$ with $\mathcal{A}_K = A$ and $\widehat{\mathcal{A}} = \mathfrak{N}$. The triple is a group object; product compatibility and full faithfulness transport this structure to $\mathcal{A}$. Raynaud's theorem makes $\mathcal{A}$ a scheme. It is smooth since its algebraic generic fibre and formal completion are smooth. If $\mathfrak{N}$ is quasi-compact, so are both parts of the triple, hence $\mathcal{A}$; together with local finite type this gives finite type.

Let X be smooth and affine of finite type over R . Full faithfulness of gluing identifies its mapping space with the space of compatible maps on the two parts:

$$\mathrm{Map}_R(X, \mathcal{A}) \simeq \mathrm{Map}_K(X_K, A) \times_{\mathrm{Map}_K(\widehat{X}_\eta, A^{\mathrm{an}})} \mathrm{Map}_R(\widehat{X}, \mathfrak{N}). \quad (39)$$

The rightmost map to the middle term is an equivalence by the formal Néron property, because $\widehat{X} \in \mathrm{SmForm}_R^{\mathrm{ad}} \subset \mathrm{SmForm}_R$. Therefore (39) reduces to

$$\mathrm{Map}_R(X, \mathcal{A}) \simeq \mathrm{Map}_K(X_K, A).$$

For an arbitrary smooth R -scheme locally of finite type, apply this equivalence on affine opens. The extensions agree on overlaps by flatness of the source and separatedness of $\mathcal{A}$, and hence glue uniquely. $\square$

Applied to the quasi-compact formal model of A^{an} for an abelian variety, this proves the algebraic assertion in Theorem C.

10. EXAMPLES AND RAMIFIED BASE CHANGE

10.1. A Tate curve. Let $q \in K^\times$ satisfy $0 < |q| < 1$, and write $n = v_K(q) > 0$ with $v_K(\pi) = 1$. For the Tate elliptic curve,

$$E_q^{\text{an}} = \mathbf{G}_{m,K}^{\text{an}}/q^{\mathbf{Z}};$$

see [17, §1]. Take $\mathfrak{P} = \text{Spf } R\langle z, z^{-1} \rangle$. Its generic fibre is the affinoid torus $|z| = 1$, and the quotient embeds it as an open subgroup of E_q^{an} . The pointed lifting property and Lemma 6.1 give the polydisc hypothesis (2). Moreover,

$$E_q(K)/\mathfrak{P}(R) = K^\times/(R^\times q^{\mathbf{Z}}) \simeq \mathbf{Z}/n\mathbf{Z}.$$

On all of SmForm_R , the formal model is the quotient

$$h_{\mathfrak{P}} \simeq \underline{(E_q(K) \times h_{\mathfrak{P}})/R^\times}.$$

Since $\mathfrak{P}_k = \mathbf{G}_{m,k}$ is connected, $\pi_0(\mathfrak{P}_k) = \mathbf{Z}/n\mathbf{Z}$.

For $0 \leq a < n$, the element π^a identifies the corresponding generic-fibre piece with the rational annulus

$$U_a = \{x \in \mathbf{G}_{m,K}^{\text{an}} : |z(x)| \leq |\pi(x)|^a \leq |z(x)|\}.$$

Each U_a maps by an open immersion to E_q^{an} , and $\mathfrak{P}_\eta \simeq \coprod_{a=0}^{n-1} U_a$. At rank-one points, these are the integral valuation classes modulo n . A point represented by z with $z^e = \pi$, $e > 1$, has nonintegral valuation class and lies outside $\mathfrak{P}_\eta$.

In the coordinate $z = 1 + X$ at the identity, the formal group law is

$$X \star Y = X + Y + XY.$$

If e denotes the tangent vector dual to dz/z at $z = 1$, then $\Lambda_{\text{disc}} = Re$. Varying n changes the component group while leaving this identity formal group unchanged.

10.2. The open additive disc and the analytic torus. On $\text{SmForm}_R^{\text{ad}}$, the open additive unit disc has a formal Néron model $\text{Spf } R\langle Y \rangle$, with generic-fibre map $T = \pi Y$; see [15, Example 1.3]. The identity map of the analytic open disc does not extend from $\mathbb{D}_R^1$ to this model, since it would require $Y = T/\pi$. This model therefore does not represent the analytic mapping functor on SmForm_R . By uniqueness on $\text{SmForm}_R^{\text{ad}}$, no other separated π -adic model can represent that functor on SmForm_R either. In this example the derivatives of pointed analytic curves form R , whereas the image of the model's Lie algebra in the tangent line with coordinate T is πR . The group is not quasi-compact, so it is outside the hypothesis of Theorem C.

For $\mathbf{G}_{m,K}^{\text{an}}$, the chart $\text{Spf } R\langle z, z^{-1} \rangle$ satisfies (2) and has index set $K^\times/R^\times \simeq \mathbf{Z}$. Theorem A gives a π -adic model representing the functor on SmForm_R , locally topologically of finite type but not quasi-compact. Gluing with the algebraic torus gives its Néron model locally of finite type; compare [2, Example 3.13].

10.3. Completions along singular centres. Let $\mathfrak{T} = \mathrm{Spf} R\langle x, y \rangle$ and let $Z \subset \mathfrak{T}_k$ be the reduced divisor $xy = 0$. Then

$$\mathfrak{X} = \mathrm{Spf}_{(\pi, xy)} \widehat{R\langle x, y \rangle}_{(\pi, xy)} \in \mathrm{SmForm}_R,$$

even though $\mathfrak{X}_{\mathrm{red}} = Z$ is singular. The model in Theorem A receives every analytic map $\mathfrak{X}_\eta \rightarrow G$. Here the auxiliary dilatation is $\mathrm{Spf}_\pi R\langle x, y, u \rangle / (xy - \pi u)$, whose special fibre maps to Z by projection.

10.4. Change of base and components. Let S/R be finite, with fraction field L , and suppose the formal models exist over both bases. The mapping property gives a canonical comparison

$$\mathfrak{N}_R(G) \widehat{\times}_R S \longrightarrow \mathfrak{N}_S(G_L).$$

For a Tate parameter with $v_K(q) = n$ and ramification index $e(S/R) = e$, the component groups are $\mathbf{Z}/n\mathbf{Z}$ and $\mathbf{Z}/(en)\mathbf{Z}$, and the comparison sends a to ea . It misses components when $e > 1$, although the multiplicative identity formal groups agree after base change.

For a chart $\mathfrak{P}$ in Theorem C, the map $\mathfrak{N} \rightarrow \underline{G(K)/\mathfrak{P}(R)}$ induces an exact sequence

$$0 \longrightarrow \pi_0(\mathfrak{P}_k) \longrightarrow \pi_0(\mathfrak{N}_k) \longrightarrow G(K)/\mathfrak{P}(R) \longrightarrow 0.$$

Here π_0 denotes the component group over k .

REFERENCES

- [1] P. Achinger, S. Gilles, and W. Nizioł, *Compactly supported p -adic pro-étale cohomology of analytic varieties*, arXiv:2501.13651.
- [2] P. Achinger and A. Youcis, *Beauville–Laszlo gluing of algebraic spaces*, J. Inst. Math. Jussieu **25** (2026), no. 2, 1193–1235.
- [3] L. Alonso Tarrío, A. Jeremías López, and M. Pérez Rodríguez, *Local structure theorems for smooth maps of formal schemes*, J. Pure Appl. Algebra **213** (2009), no. 7, 1373–1398. Preprint: arXiv:math/0605115v3.
- [4] A. Bertapelle and C. D. González-Avilés, *The Greenberg functor revisited*, revised preprint, arXiv:1311.0051.
- [5] B. Bhatt, *Lecture II: δ -rings*, prismatic cohomology lecture notes. Author’s notes.
- [6] S. Bosch, *Lectures on Formal and Rigid Geometry*, Lecture Notes in Mathematics 2105, Springer, 2014.
- [7] S. Bosch and W. Lütkebohmert, *Néron models from the rigid analytic viewpoint*, J. Reine Angew. Math. **364** (1986), 69–84.
- [8] S. Bosch, W. Lütkebohmert, and M. Raynaud, *Néron Models*, Ergebnisse der Mathematik und ihrer Grenzgebiete, third series, vol. 21, Springer, 1990.
- [9] B. Conrad, *A modern proof of Chevalley’s theorem on algebraic groups*, author’s version, 2002.
- [10] A. J. de Jong, *Crystalline Dieudonné module theory via formal and rigid geometry*, Publ. Math. IHÉS **82** (1995), 5–96.
- [11] L. Fantini and D. Turchetti, *Galois descent of semi-affinoid spaces*, Math. Z. **290** (2018), 1085–1114. arXiv:1703.03698.
- [12] U. Hartl and W. Lütkebohmert, *On rigid-analytic Picard varieties*, J. Reine Angew. Math. **528** (2000), 101–148.
- [13] R. Huber, *A generalization of formal schemes and rigid analytic varieties*, Math. Z. **217** (1994), 513–551.
- [14] R. Huber, *Étale Cohomology of Rigid Analytic Varieties and Adic Spaces*, Aspects of Mathematics E30, Vieweg, 1996.
- [15] C. Kappen, *Néron models of formally finite type*, arXiv:1106.2506.
- [16] J. N. P. Lourenço, *The Riemannian Hebbbarkeitssätze for pseudorigid spaces*, arXiv:1711.06903.
- [17] W. Lütkebohmert, *From Tate’s elliptic curve to abeloid varieties*, Pure Appl. Math. Q. **5** (2009), no. 4, 1385–1427.
- [18] A. Mayeux, T. Richarz, and M. Romagny, *Néron blowups and low-degree cohomological applications*, arXiv:2001.03597, version 4.
- [19] *From analytic discs to Néron models: Cartier curves, group smoothening, and Beauville–Laszlo gluing*, unpublished manuscript, 22 September 2026, 18 pp.
- [20] M. Raynaud, *Spécialisation du foncteur de Picard*, Publ. Math. IHÉS **38** (1970), 27–76.
- [21] The Stacks Project Authors, *The Stacks Project*. Tags 052Q, 07Z3, 0DE8, and 0GVP.
- [22] A. Vasiu, *On two theorems for flat, affine group schemes over a discrete valuation ring*, Cent. Eur. J. Math. **3** (2005), no. 1, 14–25.
- [23] T. Wedhorn, *Adic Spaces*, lecture notes, 2012.
- [24] T. Zink, *Cartiertheorie kommutativer formaler Gruppen*, with the collaboration of H. Reimann, Teubner-Texte zur Mathematik 68, Teubner, Leipzig, 1984.